

Placebo effects and emotion regulation: Conceptual and neural similarities and differences

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SUMMARY FOR THIS DRAFT

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Abstract

Placebo effects are a family of diverse effects that tap into the brain and body's latent endogenous therapeutic potential. They are driven by multiple types of psychological and brain responses to the treatment context, including memories and learned associations from past experiences, expectations about future outcomes, cognitive appraisals of the self and context, and emotional states arising from these. Likewise, these psychological and brain responses arise from multiple aspects of the therapeutic encounter, including the relationship with the care provider, specific treatment cues like procedures and setting, and suggestions. Thus, many forms of placebo effect depend on the thoughts and imagination of the patient, and their interaction with the social and physical elements of the treatment context. In this way, placebo effects may overlap with cognitive self-regulation, a family of techniques that involve using attention, appraisal, and imagination to influence one's emotions, pain, and motivation based on regulatory goals. In this chapter, we focus on the relationship between placebo effects and self-regulation of emotion. We review research on placebo effects, focusing on the idea that changes in brain processes related to affect and motivation are central and consistently identified across disorders. We then describe studies of the cognitive regulation of emotion and pain, highlighting several commonalities shared by self-regulation and placebo. The picture that is emerging is one in which appraisals of the treatment context – influenced by both placebo treatments and self-regulatory goals – shape brain systems involved in the construction of value, feelings, and motivation. These systems are central to clinical outcomes that depend on feelings and function, and interact with learning processes to alter therapeutic trajectories across disorders.

Section 1: Many routes to placebo effect

The placebo effect has long fascinated scientists and clinicians for its remarkable ability to influence a wide range of physiological and psychological outcomes. Although historically viewed as an artifact in clinical trials, placebo effects have become widely recognized as tapping into a core set of processes that shape affect and personal well-being through belief, expectation, and learned associations (Benedetti, 2020). Placebo treatments have been shown to influence a wide range of clinical outcomes across psychological and neurological disorders (Ashar et al., 2017; Benedetti, 2008). These include symptom-related outcomes that are core diagnostic features of chronic pain and mood disorders, such as pain, anxiety, depression, and emotion (Flaten et al., 2011; Schienle et al., 2013). They also include functional outcomes in neurological and substance use disorders, including movement (Rommelfanger, 2013), reward learning in Parkinson's disease (Schmidt et al., 2014), and drug craving (Koban, Kross, et al., 2017). In some cases, placebo and placebo-like effects have been reported to influence physiological outcomes as well, including autonomic responses to pain-predictive cues (Jepma et al., 2018)(Eippert et al., 2009; *Website*, n.d.-a) and subthalamic nucleus stimulation (Benedetti et al., 2004; Colloca et al., 2004; Frisaldi et al., 2014), cortisol responses to ischemic pain (Benedetti et al., 2006), and ghrelin responses to food cues (*Website*, n.d.-b). Placebo effects on symptoms are often larger than those on physiological outcomes (Meissner et al., 2007; Wechsler et al., 2011), but under the right conditions, placebo treatments appear to be able to influence peripheral physiology in ways that could compound over time to matter for various disorders.

In some cases, placebo effects can be quite durable, persisting for weeks following treatment or possibly even years. For example, placebo antidepressant pills can elicit benefits for at least eight weeks post-treatment (Rutherford & Roose, 2013), and some trials of open-label placebo (more on this below) appear to influence pain several years later (Carvalho et al., 2021). Recently, we found that a single sham injection into the back, compared with usual care without the injection, resulted in reduced pain a month post-injection, and reduced problems with sleep, depression, and anger one year later (Ashar et al., 2024; Carvalho et al., 2021).

It is important to distinguish placebo effects from placebo responses. Placebo effects are improvements in outcomes thought to be caused by the placebo, and by extension, caused by the patient's internal processing of the treatment context. Placebo responses, by contrast, are a more inclusive category of phenomena that also encompasses regression to the mean, sampling bias, elevation bias, and other types of statistical artifacts (Kienle & Kiene, 1997; Wager & Fields, 2013).

Most randomized clinical trials can provide information on the magnitude of the placebo response, but are unfortunately not suitable for studying placebo effects. Identifying placebo effects generally requires comparing a group receiving a placebo intervention with a control group that does not – for example, a natural history or no-treatment group. For example, in a parallel groups design, a placebo group may be administered a sham drug with suggestions that it is an effective verum treatment, and a control group is given the same sham drug but without these positive suggestions. The same design can be used to study placebo effects related to other treatments, including neurostimulation, surgery, and more.

The parallel groups design is not the only way to study placebo effects. In a within-person crossover design, each study participant is tested in both placebo and control conditions, usually with treatment order counterbalanced to avoid confounding placebo effects with natural history. This design is most common in the neuroimaging studies we discuss below (Figure 2A). The open-hidden design is another effective choice, in which participants are given a verum treatment under two conditions: An open-label condition in which they know they are receiving it, and a hidden condition in which they do not (Colloca et al., 2004). Like the parallel groups and crossover placebo designs, these designs allow for the effects of the patients' knowledge and expectations about the treatment to be isolated and studied.

Using these types of designs to isolate placebo effects is essential for understanding the brain's and body's ability to mount endogenous therapeutic responses, and how various aspects of the treatment context can tap into a person's latent healing potential. For example, a meta-analysis in 2008 showed that placebo responses in Parkinson's disease are large and growing over time (Goetz et al., 2008). By this the authors mean that patients in placebo groups tend to improve quite a bit over time. However, it is not clear how much of the overall placebo response is caused by regression to the mean and related phenomena. For example, participants may enroll in treatments when their symptoms are high and appear to improve simply due to natural symptom fluctuation. In addition, there is an initial elevation bias in symptom reports when entering many kinds of studies, which disappears over the course of several weeks following enrollment (Shrout et al., 2018). Additionally, it is not clear how much of the growing effect across years is due to drift in diagnostic standards across time. If patients with less severe disease are increasingly included in trials, spontaneous improvement on placebo will thus tend to grow over time. Thus, it is not clear whether the placebo responses documented in Goetz et al. 2008 reflect any endogenous capacity for functional improvement in Parkinson's disease.

Experimental studies of placebo effects can address this shortcoming. For example, a seminal crossover trial by Fuente-Fernandez et al. (de la Fuente-Fernández et al., 2001) studied Parkinson's patients off medication and on sham apomorphine. They found enhanced dopaminergic activity (reduced D2 dopamine receptor binding) in the striatum, isolating a placebo effect on endogenous dopamine responses. Similar effects on dopamine have been found with a placebo painkiller (Scott et al., 2008), and effects on other aspects of Parkinson's disease have been found using similar designs, including those on motor function and associated subthalamic neural responses (Benedetti et al., 2004) and reward learning and associated ventral striatal prediction errors (Schmidt et al., 2014). These experimental studies suggest that at least some portion of the large placebo responses observed in Goetz et al. 2008 is due to placebo effects, improvements triggered by patients' beliefs that they are receiving help and the shifts in psychological processes that accompany them.

Initial theoretical models of placebo effects emphasized two kinds of mediating psychological processes: Conditioning and expectations (Bingel, 2013; Kirsch, 1999; Voudouris, 1990). Conditioning is a family of procedures involving pairing of context cues with affectively charged stimuli, stimuli that elicit *unconditioned responses* like escape, protective behaviors, avoidance, pursuit, and consumption. The placebo treatment is thought to consist in part of a set of context cues, including the look and feel of a needle used for a therapeutic injection, the clinical setting, and interactions with healthcare providers. Pairing placebo treatments with positive outcomes, such as pain relief or symptom improvement, can condition individuals to consciously or unconsciously associate these contextual cues with therapeutic benefits. Over time, these cues alone may elicit conditioned responses, such as reduced pain perception or physiological changes, even in the absence of active medication.

The conscious aspects of conditioning are related to expectations, beliefs about future events. In some cases, conditioning can elicit learned conditioned responses mediated by neuroplasticity in specific neural pathways, such as intra-spinal pathways for conditioning of shock-induced motor responses (Grau et al., 2022) or pathways through the basolateral amygdala for conditioned threat responses to auditory tones. However, the power of conditioned cues to elicit responses lies in large part on their information value, and thus their ability to elicit conscious expectations (Rescorla, 1988). In addition, expectations are flexible cognitive processes that can be elicited in humans in many other ways as well, including verbal suggestions, memories of prior experiences, and social observation of others (Koban, Jepma, et al., 2017). These interact with one's personal predispositions and cultural beliefs to influence how patients anticipate and evaluate the effectiveness of treatments. For example, verbal instructions about the monetary value of a treatment can modulate both pain experience and pain-related activity in the brain and spinal cord

(Tinnermann et al., 2017). Conditioning and expectations are not mutually exclusive; rather, the process of conditioning (pairing of cues and unconditioned stimuli) elicits both unconscious learned associations and conscious expectancies, both of which shape real-time cognitive appraisals of the treatment context (See Figure 1A).

Traditionally, the placebo effect was thought to rely on deception—for example, patients being led to falsely believe that placebo pills, which are inert, are effective in treating their pain. However, emerging research has revealed that the placebo effect can also be achieved through other routes, including engaging the psychosocial context surrounding treatment without deception. These include effects of supportive care and open-label placebos, placebo treatments delivered with explicit instructions that the treatment is an inert substance with no active ingredients (Carvalho et al., 2016; Kaptchuk et al., 2010; Locher et al., 2017) or even appealing to patients' goals (e.g., explicit emotion regulation and self-management strategies) (Braunstein et al., 2017).

These new approaches have demonstrated comparable effectiveness in relieving patients' affective distress and pain experiences. The challenge for the field, however, is to understand which types of interventions have an impact at various levels and which produce lasting effects over time. For example, it is clear that open-label placebos can influence patients' self-reports of symptoms in some cases (Charlesworth et al., 2017), even over long time scales of a year or more (Ashar et al., 2024; Carvalho et al., 2021). However, open-label placebos and other forms of suggestion can often impact subjective symptom assessments without affecting objective measures of function, physiology and cognitive performance (Buhle et al., 2012; Hartmann et al., 2023; Kleine-Borgmann et al., 2019; Wechsler et al., 2011) (for review see (Meissner et al., 2007)), and the longevity of subjective improvements is variable (Ashar et al., 2024; Kleine-Borgmann et al., 2023).

A central premise of this chapter is that the strongest pathways to the placebo effect primarily involve affective and motivational processes, including subjective assessments of harms, benefits, personal capacities, and future implications. Accordingly, the most robust placebo effects are found on brain systems related to these processes and outcomes that depend on them. We refer to these meaning-making processes under the umbrella term 'appraisals'. Appraisals are crucial ingredients in the construction of emotion (Barrett et al., 2015; Ellsworth, 2024) and in decision-making, and are thus essential determinants of health-promoting and health-reducing behavior. Thus, there can be both direct effects of placebo-induced appraisals on symptom experience and indirect effects mediated by changes in behavior (Figure 1B).

An important corollary of this view is that the same central affective and motivational processes can be influenced both by placebo treatments and by instructed emotion-regulation strategies. Thus, in addition to effects of supportive care, suggestion, and open-label placebos, care providers can tap into the “power of placebo” by working with patients to give them explicit strategies for self-regulation. In addition, comparing the neural mechanisms and systems involved in placebo effects and emotion regulation may help illuminate common principles and effects shared by both. The remainder of this chapter explores the commonalities and differences between neural effects of placebo and instructed emotion regulation.

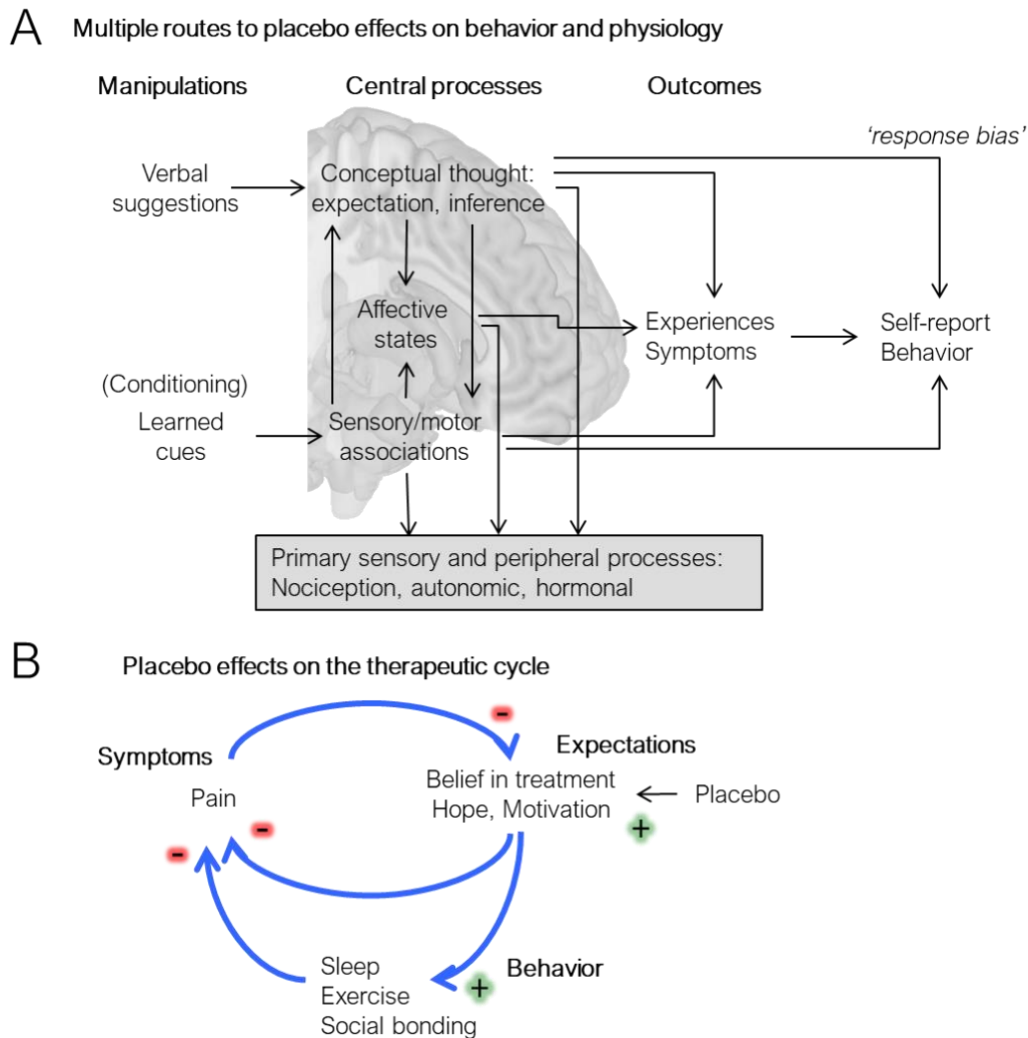


Figure 1. Theoretical model showing many routes to placebo effects. **A)** Placebo treatments can tap into (1) unconditioned predispositions and learned associations, (2) affective and emotional states, and (3) conceptual appraisals, including expectations about the future and inferences about the causes of events. These can influence multiple outcomes, at both the experiential levels (e.g., pain) and physiological levels (e.g. autonomic activity). In some cases, effects may be attributable to response bias, which we define here as a direct effect of placebo suggestions

*on self-report in the absence of a change in symptom experience or physiology. But there are many pathways by which placebo treatments can produce more clinically meaningful effects as well. **B)** Placebo treatments, like other psychosocial interventions, inject new input into ongoing feedback cycles of symptom experience, thought, and behavior that occur throughout the process of experiencing and recovering from any disorder. Here, we illustrate these feedback cycles using pain as an example. Pain experience reduces positive appraisals, including hope, motivation, and belief in treatment efficacy. These positive appraisals naturally antagonize pain, creating a negative feedback cycle. In such a system, the trajectory of pain over time depends on the balance between these mutually antagonistic effects. Placebo treatments are designed to enhance positive appraisals, and may tip the balance towards recovery over time in some cases. They can also positively influence health behaviors, such as decisions favoring physical activity, social bonding, and self-care (e.g., sleep hygiene), with further benefits on symptom trajectories over time.*

Section 2: Placebo effects in emotional and motivational processes: An appraisal account

Placebo effects extend beyond their role in symptom relief to fundamentally shape emotional and motivational processes, acting through core appraisal mechanisms that influence how individuals interpret their experiences and anticipate future outcomes. Ashar, Chang, and Wager (2017) propose that placebo effects operate primarily by engaging cognitive appraisals, which are flexible, self-referential evaluations of the meaning and significance of events. These appraisals integrate contextual cues, prior experiences, and expectations, enabling placebo treatments to reframe emotional responses and motivational drives. By this view, the placebo effect is in large part a 'meaning response' (Brody, 2018; Moerman & Jonas, 2002), an active reconceptualization of one's symptoms and abilities and what they mean for the self and the future.

The understanding of the connection between cognitive appraisals and placebo effects has been significantly advanced by neuroimaging research. A series of studies have revealed how placebo interventions alter neural activity, reshaping emotional and anticipatory states by engaging specific brain networks. Placebo analgesia is the best-studied type of placebo effect when it comes to brain mechanisms, and thus offers some useful general lessons. Placebo treatments have been shown to reduce pain-related activity in regions associated with the 'ventral attention network' (VAN; sometimes referred to as the 'salience network' or 'cingulo-opercular network'), a functional system that is jointly important for the construction of symptoms such as pain, associated decision-making, and the regulation of autonomic responses as well. Key nodes in this network are both activated by painful stimulation and down-regulated by placebo analgesic treatments, including the mid-insula, anterior midcingulate (aMCC), and medial thalamus (Ashar

et al., 2017; Wager & Atlas, 2015; Zunhammer et al., 2021). In addition, placebo effects on pain-related activity also extend to key regions within the ‘somatosensory network’ for afferent nociceptive signaling and pain construction, including posterior insula and ventrolateral thalamus, at least in some cases (Zunhammer et al., 2021).

Placebo effects on nociception- and pain-specific brain measures are rare. Instead, the most common effects of placebo painkillers are in regions that mediate many types of affective and decision-making processes. In addition to VAN-associated regions that are important for shaping action and action policy, placebo treatments frequently influence regions associated with the ‘default mode network’ (DMN), which tracks subjective value and self-relevance and, in coordination with ‘limbic’ regions such as the ventral striatum and cingulate and brainstem centers such as the periaqueductal gray, mediates multiple types of self-referential decision-making processes (Koban et al., 2021; Roy et al., 2012).

This principle was recently tested and demonstrated in a recent large-scale fMRI study of pain and placebo analgesia with pre-registered brain outcomes (Botvinik-Nezer et al., 2024). This study tested two kinds of placebo effects on brain activity and pain reports: A conditioned placebo effect on heat pain, and an unconditioned transfer of placebo effects to a new pain type, mechanical pain. Placebo effects on pain reports were strong for both types of placebo, with standardized mean difference effect sizes of $d = 0.6$, comparable to previous meta-analyses. In addition, robust pain-related activity was identified in expected regions and multivariate neuromarkers, validating the quality of the imaging data. However, the study found that placebo treatment did not reduce activity in any of the pre-specified regions of interest associated with nociceptive pain, such as the posterior insula, and in fact significantly *increased* pain-related activity in some of these regions in the transfer condition. In addition, placebo treatment had no effect on a validated neuromarker related to nociceptive pain, the Neurologic Pain Signature (Wager et al., 2013), confirming similarly small effects of placebo in a previous pre-registered meta-analysis (Zunhammer et al., 2018). However, both the conditioned placebo treatment and the transfer placebo test significantly reduced pain-related activity in a second pain neuromarker associated with evaluative aspects of pain (Woo et al., 2017). This neuromarker was trained and validated to track cognitive influences on pain perception independent of stimulus intensity and Neurologic Pain Signature responses, and includes activity in regions in the DMN and beyond associated with value, motivation, and decision-making – including the ventromedial prefrontal cortex, nucleus accumbens, and dorsolateral prefrontal cortex.

These findings in pain are paralleled by converging evidence showing placebo effects on many of the same value- and decision-making associated brain regions in different disorders. One of the

most consistent findings in the placebo analgesia literature is decreases in the aMCC and mid/anterior insula (Atlas & Wager, 2012), ‘hub’ regions associated with the VAN that likely play broad roles in assessing relevance and shaping behavioral responses across a wide range of affective and cognitive states. Similar placebo-induced decreases have been found in other affective challenges as well, including itch (van de Sand et al., 2018) and tussis (Leech et al., 2013).

Effects on medial fronto-striatal systems extend beyond pain as well. In Parkinson’s disease, expectations about dopamine release induced by placebo interventions have been found to have comparable effect with real dopaminergic drugs (mainly Cinemet, a combination of levodopa and carbidopa) facilitating reward learning process in VMPFC and ventral striatum (Schmidt et al., 2014). Similarly, in patients with major depression, placebo effects have been shown to improve treatment outcomes by elevating μ -opioid neurotransmission in the ventromedial prefrontal cortex (specifically, subgenual cingulate), nucleus accumbens/ventral striatum, and amygdala (Peciña et al., 2015). Increased opioid activity also predicted better antidepressant response and greater symptom improvement (Peciña et al., 2015). In studies on substance misuse, placebo drugs (alcohol, caffeine, and amphetamine) have been found to reduce craving and produce subjective drug experiences, and in some cases affect cognitive performance measures such as reaction time (Galindo et al., 2020; Volkow et al., 2003). In the brain, expectations of nicotine consumption increased monetary reward-related and reward prediction error-related activity in the nucleus accumbens/ventral striatum, and increased reward effects on future decisions in an investment task (Gu et al., 2015). Beliefs that one is receiving alcohol and methylphenidate have also been shown to reduce striatal dopamine (Yoder et al., 2009) and activity in orbitofrontal cortex (Volkow et al., 2003), consistent with effects on prediction error signaling. Placebo treatments can modulate hunger and food craving as well. In one study, consuming a placebo drink that was suggested to up- or down-regulate hunger modulated subjective experience accordingly – and suggestions of decreased vs. increased hunger down-regulated responses to food cues in ventromedial prefrontal and posterior cingulate cortices (Khalid et al., 2024), DMN-associated regions related to valuation and value-based influences on behavior. Further analyses demonstrated that placebo suggestions influenced the degree to which these circuits encode food value and how strongly participants considered tastiness in their food choices. These findings suggest that a key function of placebo interventions is their ability to engage neural systems associated with value, value learning, and related affective states, producing effects on symptoms and decision-making that often parallel those of verum treatments.

In addition to effects on systems related to affective value, lateral frontoparietal systems, and the dorsolateral prefrontal cortex (DLPFC) in particular, may play a key role in maintaining context

information related to placebo treatment and modulating activity in affective processing systems accordingly. The DLPFC has shown consistent activation in expectation-related placebo analgesia (Wager et al., 2005; Zubieta et al., 2005), and is thought to support expectancy-driven modulation of pain (Benedetti et al., 2005; Wager et al., 2004), potentially by activating descending pain-inhibition pathways in conjunction with medial DMN-related systems (Bingel et al., 2006) in some cases or by directly modulating the construction of value. Functional imaging studies have shown that DLPFC becomes active when individuals anticipate relief from pain (Benedetti, 2009; Wager et al., 2004), and the degree of anticipatory activation correlates with the magnitude of placebo analgesia (Wager et al., 2011). DLPFC is also thought to regulate the aMCC, which plays a primary role in integrating the sensory, cognitive, and affective dimensions of pain (Craggs et al., 2007) and determining action policy (Johansen et al., 2001; Roy et al., 2014). This modulatory role has been further confirmed by neural stimulation studies. Studies have shown that potentiating DLPFC using transcranial direct current stimulation (TDCS) can facilitate placebo effects (Tu et al., 2021), while a transient disruption of DLPFC using repetitive transcranial magnetic stimulation (r-TMS) activity blocked placebo analgesia (Krummenacher et al., 2010). Similar modulation has also been reported in a social pain scenario (Wang et al., 2023). DLPFC plays a role in placebo effects beyond pain as well, along with representing context across many different tasks (Miller & Cohen, 2001). For example, in the food valuation study above, placebo effects on the valuation of food were mediated by activity in DLPFC (Khalid et al., 2024). Taken together, these findings suggest DLPFC's role in maintaining expectations and thereby influencing value and symptom construction.

Together, these findings underscore the transformative potential of placebo treatments to influence emotional and motivational processes across multiple types of symptoms and task contexts. By engaging brain systems involved in appraisal, reward, and emotion regulation, placebo effects provide a unique window into understanding the interplay between belief, context, and neural activity.

Section 3: Emotion regulation: another path to influence cognitive appraisal

One of the key features of the placebo effect is its impact on affective and motivational systems, including those mentioned above. Alteration of these systems converges with effects of other ways of influencing appraisals, including the classic approach of instructed emotion regulation. Emotion regulation paradigms use instructions that parallel those delivered in Cognitive Behavioral Therapy and related treatments for mental health disorders, instructing study participants to alter their emotional responses by changing the meaning context, altering appraisals about the nature of stimuli (e.g., their harmfulness to oneself or another) and future implications. In this section, we introduce the foundational concepts of emotion regulation, its

neural mechanisms, and its clinical applications. In the following section, we compare it with the placebo effect in both conceptual and neural level.

Explicit emotion regulation, often referred to as self-directed emotion management, is a widely used approach in clinical practice for addressing affective disorders. Its mechanisms and outcomes align closely with those of the placebo effect, making it a meaningful comparison point for understanding how placebo influences affective experiences. A prominent example is cognitive restructuring in Cognitive Behavioral Therapy (CBT) (Hofmann et al., 2012), one of the most commonly employed interventions for Major Depressive Disorder (MDD) and related mental health disorders. At its core, CBT incorporates various emotion regulation strategies, encouraging patients to directly target their emotions and modify specific stages of the emotion generation process (Beck, 2020; Garke et al., 2024).

In a widely recognized process model of emotion regulation (Gross, 2011), emotions emerge through a series of stages: a person encounters an emotionally relevant situation, attends to it, appraises its significance, and ultimately experiences an emotional response. Emotion regulation strategies intervene at one or more of these stages to shape emotional outcomes. For instance, distraction alters situation selection, cognitive reappraisal modifies the valuation process, and expressive suppression modulates the display of emotion. These strategies are unified by their reliance on cognitive processes related to prefrontal cortical function, which in turn is thought to influence brain systems involved in affect and emotion generation. Among these strategies, cognitive reappraisal is considered particularly important, as it is thought to change the nature of an emotional response to a stimulus (Ochsner & Gross, 2005). Situation selection and attention also play important roles, but they are forms of avoidance that can incur behavioral costs when relied on too heavily, whereas expressive suppression is a form of masking emotion thought be costly in that it can produce physiological hyperarousal (Richards & Gross, 1999). The change in emotion construction afforded by reappraisal underpins its use in clinical treatments for affective disorders.

The neural mechanisms of emotion regulation involve a broad network of brain regions spanning cortical and subcortical areas. This "emotion regulation network" has been consistently identified in meta-analyses of neuroimaging studies across various regulation strategies (Buhle et al., 2014; Dörfel et al., 2014; Morawetz et al., 2017). These analyses have revealed robust patterns of activation that differentiate healthy individuals from populations with mood disorders (Picó-Pérez et al., 2017). The network's core regions can be categorized into three functional components, each playing a distinct role in emotion regulation.

The first component, the frontal-parietal network, includes the dorsolateral prefrontal cortex (DLPFC), ventrolateral prefrontal cortex (VLPFC), inferior parietal cortex (IPC), and temporal

parietal junction (TPJ). This network is central to cognitive control processes such as goal maintenance, selective attention, and response inhibition. The second component, the midline brain system, encompasses the ventromedial prefrontal cortex (VMPFC), dorsomedial prefrontal cortex (DMPFC), supplementary motor area (SMA), pre-SMA, dorsal anterior cingulate cortex (dACC), and posterior cingulate cortex (PCC). The VMPFC and PCC are key hubs in the DMN mentioned above. These regions are primarily involved in appraisal and valuation, assigning emotional significance to stimuli. Lastly, the lateral temporal cortex contributes to memory retrieval and the integration of past experiences, which are essential for effective emotion regulation.

An emerging area of research involves the use of neural stimulation to enhance the functioning of these cognitive control regions, with significant implications for clinical practice (Feesser et al., 2014; He et al., 2021; Zhao et al., 2021). For example, transcranial direct current stimulation (tDCS) and transcranial magnetic stimulation (TMS) targeting the DLPFC and VLPFC have shown promise in reducing negative affect. Studies have demonstrated that stimulation of these regions enhances the effectiveness of emotion regulation strategies, making it a promising avenue for therapeutic intervention.

While the regions engaged during emotion regulation are well-characterized, the areas targeted for downregulation are more variable across strategies and studies. The amygdala is the most consistently downregulated region (Buhle et al., 2014), though this effect depends on the specific strategy employed. Cognitive reappraisal, for instance, has been found to exert relatively weak effects on the amygdala (Bo et al., 2024; Morawetz et al., 2017; Uchida et al., 2015). As sample sizes have grown and evidence accumulated across studies, it has become increasingly clear that reappraisal of negative emotion involves changes that extend beyond amygdala, with changes in emotion potentially directly mediated by changes in cortical systems (Bo et al., 2024). Experiencing negative affect involves increased activation of sensory modality-specific perceptual pathways (Čeko et al., 2022; Satpute et al., 2015) – e.g., visual thalamocortical pathways for visual stimuli, auditory thalamocortical pathways for auditory stimuli – and activity in these pathways predicts the degree of negative affect experienced (Čeko et al., 2022). Reappraisal has most commonly been studied in relation to aversive visual stimuli, and successful downregulation of negative emotion in this context is related to reductions in stimulus-evoked responses in regions important for high-level visual processing and semantic meaning, along with activation of medial and lateral fronto-parietal systems involved in generating appraisals (Bo et al., 2024). These findings underscore the complexity of emotion regulation, which relies on a dynamic interplay between activating regulatory networks and suppressing emotion-generative regions, and suggest that like placebo effects, much of the change in emotion might be directly related to

alterations in association cortices rather than involving suppression of subcortical affective responses.

In summary, emotion regulation involves a sophisticated network of cortical and subcortical regions that support cognitive control, appraisal, and memory retrieval processes. These mechanisms are not only effective in modulating affective experiences but also provide a valuable reference point for understanding the affective modulation induced by placebo interventions.

Section 4: Key similarities and differences between placebo effect and emotion regulation in neurobiology

As reviewed in the previous sections, both the placebo effect and emotion regulation target the appraisal processes underlying affective experiences and are capable of shaping subjective experiences. Both processes influence how stimuli or situations are valued, determining whether an event or treatment is appraised as threatening, beneficial, or neutral.

Despite these similarities, the two processes exhibit clear differences. Emotion regulation is typically a deliberate, self-initiated process that involves conscious awareness and is directed toward a clear emotional goal. In contrast, the placebo effect does not always require conscious awareness of the true situation or the explicit intention to influence emotions (with the exception of open-label placebo interventions). Thus, placebo effects influence appraisals in an implicit manner, whereas emotion regulation influences appraisals in an explicit manner shaped by the patient's goals. Furthermore, the temporal dynamics of the two processes differ. The placebo effect often unfolds over a longer time frame, beginning as early as verbal instructions are provided and continuing as the belief in the treatment strengthens throughout the procedure. Emotion regulation, on the other hand, is usually initiated voluntarily by the individual and is terminated when effort is no longer applied.

Given these links between the placebo effect and emotion regulation, an overview of the comparison between the brain systems underlying them might be helpful to further understand the placebo effect.

The systems activated by cognitive reappraisal have been highlighted in several meta-analyses (Buhle et al., 2014; Morawetz et al., 2017; Zunhammer et al., 2021). Key regions, including the DLPFC and VMPFC, are frequently activated in both reappraisal and placebo paradigms. However, these regions are not only large but also highly complex, encompassing subregions with distinct functions. For example, the DLPFC spans a large area of the cortex, with different functions localized to its anterior and posterior parts (Badre, 2008). To address this limitation, this section integrates data from large-scale studies, presenting activation maps for the placebo effect

(Zunhammer et al., 2021) (N = 603) and emotion regulation (Bo et al., 2024) (N = 358) side by side. These maps are aligned within a unified brain template, enabling a more detailed and fine-grained comparison of the neural mechanisms underlying these processes (Figure 2).

The direct comparison in Figure 2 highlights several notable distinctions and areas of overlap between the brain systems engaged by placebo effects and emotion regulation. The most intuitive difference lies in the effect size: the placebo effect demonstrates a much smaller neural effect size compared to emotion regulation. However, this disparity may not accurately reflect the true brain activation associated with these processes. The placebo effect meta-analysis encompasses a broad range of stimuli and experimental designs, whereas the emotion regulation activation map is derived from two datasets using the same regulation strategy and similar experimental paradigms, making the latter more uniform.

Another factor contributing to this difference is the temporal nature of the placebo effect. For instance, in typical placebo paradigms (e.g. (Botvinik-Nezer et al., 2024; Schafer et al., 2015), participants are exposed to placebo-related cues (e.g., applying a cream or receiving verbal instructions) well before the actual stimulus (e.g., heat pain). Brain responses related to placebo effects are typically measured during the stimulus period, but the brain mechanisms initiating the placebo response may already be active during the cue phase and/or before any test stimuli are presented. This early activation could reduce the size of the responses to the test stimuli, as placebo treatment may establish a preparatory set that obviates the need to engage relevant appraisal systems during stimulus processing. In contrast, in emotion regulation paradigms, the regulatory goal is often set only seconds before the test stimulus appears, requiring substantial recruitment of appraisal-related brain systems while the test stimulus is presented. In variants of emotion regulation that establish a sustained preparatory set – e.g., adopting an ‘acceptance mindset’ and applying it to a series of test stimuli – down-regulation of emotion appears to be achieved without activation of fronto-parietal appraisal systems (Gard et al., 2012; Kober et al., 2019). In summary, while both placebo effects and emotion regulation share a similar overarching goal, they might differ in their temporal and intensity profiles. Emotion regulation often elicits a larger, more immediate brain response due to its clear regulatory objective and need for alterations in appraisal in response to each test stimulus, whereas placebo effects reduce negative affect without engaging appraisal systems to the same degree at the time of the test stimulus, possibly by preparing activity in appraisal networks in advance and sustaining a preparatory set over time.

In spite of these differences, one of the most prominent similarities between the two processes is the recruitment of the DLPFC and superior parietal cortices. These regions are commonly co-

activated in a wide variety of cognitive control tasks as well as emotion regulation tasks, and are associated with the ‘fronto-parietal’ network. Almost the entire DLPFC is robustly activated by emotion regulation, while the placebo effect shows weaker but distinct clusters across the anterior, middle, and posterior DLPFC. DLPFC is traditionally considered a region of cognitive control for affect manipulation (Ochsner et al., 2012). However, recent studies began to propose that the DLPFC plays a broader role in value construction rather than merely suppressing affective instincts (Hutcherson & Tusche, 2022; Wilson et al., 2023). In these studies, DLPFC activity has been shown to increase as value-related evidence is accumulated to support both affective (e.g., hedonic) and regulatory goals. In the context of emotion regulation and placebo effects, this may reflect a shared process of updating evidence for cognitive appraisals of stimuli, ultimately altering the associated affective response.

Interestingly, studies on open-label placebo interventions also report an absence of lateral prefrontal activation, suggesting that cognitive appraisal may play a less prominent role in these paradigms (Ashar et al., 2024; Schaefer et al., 2022), though they may engage VMPFC-subcortical pathways associated with value and meaning (Ashar et al., 2024). However, the study of the neural mechanisms of open-label placebo is in its infancy, and more direct comparisons with deceptive placebo paradigms and other regulatory strategies are needed.

Beyond the DLPFC, other overlapping regions include the precuneus and superior temporal sulcus, both of which are involved in the integration of multisensory information and emotion appraisal (Beauchamp, 2005; Buckner et al., 2008; Calvert et al., 2000; Utevsky et al., 2014). The precuneus is also implicated in self-referential processes (Cavanna & Trimble, 2006; Northoff et al., 2006), potentially linking the integration of sensory information with personal relevance and emotional impact, and with control of working memory (Wager & Smith, 2003), providing a link to other forms of cognitive control. Whereas the ventral precuneus and PCC are associated with the DMN, the dorsal portion of precuneus activated in common in emotion regulation and placebo paradigms is part of the fronto-parietal network, supporting the idea that both processes may engage networks associated with context-based cognitive control.

Together, these findings suggest that both the placebo effect and emotion regulation rely on the engagement of systems involved in the appraisal of affective meaning, which overlap with networks involved in the control of attention and working memory more broadly. These networks may help maintain information about the context of stimulation, redirect emphasis towards different qualities of the stimulus, and imagine different future implications for harm. These evaluations are modulated either by explicit goals (emotion regulation), or by preset expectations (placebo effect). Ultimately, both processes downregulate negative affect and enhance overall well-being.

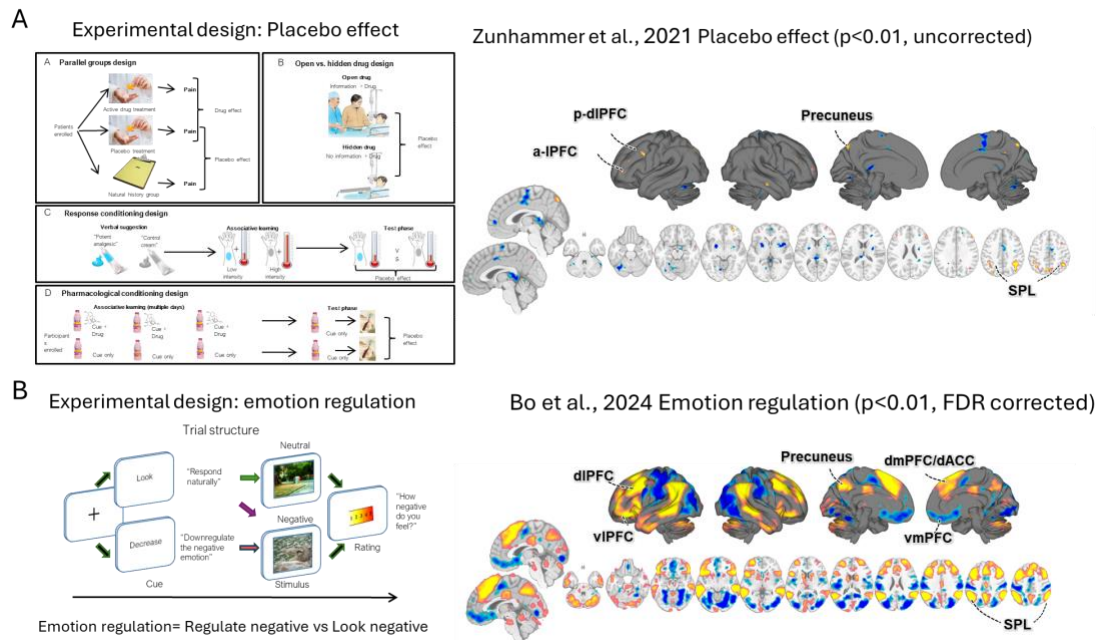


Figure 2. A comparison of brain activation patterns during placebo analgesia and instructed regulation of negative emotional responses to aversive visual stimuli. A) Left: Three common designs for eliciting placebo effects. The studies on which the present results are based (Zunhammer et al. 2021) mainly use the ‘response conditioning’ design. Right: Brain activity differences for placebo and control conditions in response to painful stimuli, from Zunhammer et al. (2021), where the placebo effect is modeled as a fixed effect. Activation maps were thresholded at $p < 0.01$, uncorrected for multiple comparisons, with a minimum cluster size of 10 voxels. B) Left: The emotion regulation paradigm used in two studies on which the present results are based (Bo et al. 2024). Right: Brain activity differences for reappraisal of negative emotional pictures vs. natural responses to matched negative pictures, from Bo et al. (2024). Statistical thresholds were set at $p < 0.01$ FDR-corrected for multiple comparisons, with a minimum cluster size of 10 voxels. Note that the data source for the placebo effect is a meta-analysis of multiple studies, while the data for emotion regulation is derived from two datasets within a single study.

Section 5: potential clinical applications of placebo facilitating emotion regulation and mental health.

Both the placebo effect and emotion regulation aim to reduce negative emotions and promote mental well-being. Despite their shared goal, each process has limitations that can hinder its effectiveness. A major challenge in emotion regulation is that it requires deliberate effort and motivation, which patients often lack, particularly during periods of heightened emotional distress. Effective strategies like cognitive reappraisal can be cognitively demanding, leaving individuals overwhelmed or unable to engage consistently. In contrast, the placebo effect relies heavily on belief and expectation; when these are weak or absent, its ability to modulate emotional responses is significantly reduced.

These challenges highlight the need for innovative approaches that leverage the strengths of both processes while compensating for their respective weaknesses. Evidence suggests that placebo effects can serve as a complementary mechanism to emotion regulation. For instance, placebo effects effectively reduce distress in response to negative emotional stimuli, such as distressing images, sad movie clips, or anxiety-inducing tasks (Glombiewski et al., 2021; Meyer et al., 2015, 2019; Rebstock et al., 2020). Unlike emotion regulation, placebo responses are largely automatic, driven by expectancy and learned associations. This automaticity makes placebo effects particularly useful in situations where cognitive resources are limited, such as under stress or fatigue (Grillon et al., 2015; Raio et al., 2013). In such contexts, placebo effects provide a low-effort means of emotional modulation that can complement more cognitively demanding regulatory strategies.

A closer examination of the interaction between placebo effects and specific emotion regulation strategies reveals opportunities for synergy. For example, cognitive reappraisal may synergize with effects of placebo in the DLPFC, supporting positive reinterpretations of stimuli and their contexts (Van Der Meulen et al., 2017; Zhang et al., 2013). It is easier to generate appraisals that are compatible with one's beliefs than appraisals that are incompatible with them – and, indeed, it may not be possible to self-generate appraisals that one does not believe in. In this way, placebo treatments could induce positive interpretations of the context, such as the idea that one is being helped and cared for and that one's symptoms will decrease over time, and explicit reappraisal instructions could help patients focus on and enhance this interpretation. Although open-label placebo treatments often elicit weaker responses than reappraisal, they still engage medial prefrontal and striatal regions, which link affective stimuli to behavior and support reward- and punishment-driven learning, and can reinforce sustained regulatory efforts (Schienle et al., 2023). Similarly, placebo treatments may enhance the efficacy of distraction strategies, which focus attention away from a painful or negative emotional stimulus. One previous study found that both placebo and distraction rely on endogenous opioid mechanisms, as both effects are reduced by the opioid antagonist naloxone (Sprenger et al., 2012). However, studies that have independently manipulated both placebo and distraction with a concurrent cognitive task have found additive effects of both, implying that both can be used side-by-side but they do not synergize (Buhle et al., 2012; Shafir et al., 2023). Potential interactions between manipulations of belief and emotion-regulation strategies need further exploration, as such interactions and their mechanisms may vary across contexts and implementations.

The clinical implications of integrating placebo effects with emotion regulation strategies are significant, particularly for individuals with mood disorders. Research suggests that the success of antidepressant medications is partially mediated by placebo mechanisms (Kirsch, 2013). Neuroimaging studies reveal overlapping activation patterns in brain regions targeted by

antidepressants and placebo responses, indicating shared pathways for symptom alleviation (Burke et al., 2022). For individuals with major depression, even the expectation of receiving an antidepressant—even when it is a placebo—can significantly reduce sadness following negative stimuli (Haas et al., 2020) and clinical symptoms of depression (Rutherford & Roose, 2013; Zilcha-Mano et al., 2019), demonstrating the powerful role of expectancy in shaping emotional responses. At a neural level, antidepressant expectancies can reduce responses to sad faces in depressed patients in the amygdala, insula, somatosensory, and temporal cortices (Zilcha-Mano et al., 2019).

Building on these findings, integrative interventions could combine placebo-like elements with emotion regulation training to address their respective limitations. For example, involvement of placebo cues, such as verbal encouragement or symbolic rituals, could enhance motivation, priming patients for more effective engagement with emotion regulation strategies. Conversely, teaching emotion regulation techniques in placebo-related contexts might amplify and sustain placebo-induced benefits, creating a positive feedback loop that fosters emotional resilience over time. Such approaches could be particularly beneficial for individuals who struggle to initiate or sustain self-regulation efforts, offering a scalable and cost-effective alternative to pharmacological treatments.

In summary, integrating placebo effects with emotion regulation strategies offers a promising avenue for advancing care across multiple symptoms and disorders, including mental health disorders, substance use disorders, and pain conditions. By addressing motivational and cognitive barriers and leveraging expectancy-driven mechanisms, this combined approach has the potential to improve emotional well-being and enhance the overall effectiveness of pharmacological, neurostimulation, and behavioral treatments. As a final thought, this integration underscores the broader importance of understanding the interplay between belief, cognition, and emotion in shaping mental health interventions—a perspective that may pave the way for more holistic and personalized therapeutic approaches.

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